MVA DLMI 2026 - Histopathology OOD Classification

Team: TODO
Jean-Vincent Martini[1,2] JEAN-VINCENT.MARTINI@ENS-PARIS-SACLAY.FR
Ayoub El Kbadi[1,2] AYOUB.KBDI@GMAIL.COM
[1] *CentraleSupélec*
[2] *ENS Paris-Saclay*

Abstract

This is a great paper and it has a concise abstract. The full code is available at https://github.com/JeanJNMV/MVA-DLMI_Histopathology_Challenge.
Keywords: List of keywords, comma separated.

1. Introduction and Problem Definition

Computational pathology has emerged as a promising application of deep learning, with transformer-based models achieving expert-level performance on a variety of tissue classification tasks. In clinical practice, however, models are often trained on data from a limited number of institutions and subsequently deployed in hospitals where staining protocols, scanners, and tissue preparation procedures differ substantially. This creates a domain shift, a discrepancy between the training distribution and the target distribution, that can cause a significant drop in classification accuracy, even when the underlying tissue morphology is similar.

This challenge addresses precisely this problem. The task is the binary classification of patches that must be labeled as either tumor or non-tumor. The training set gathers images from three different medical centers, while the validation and test sets originate from two additional, unseen centers. The appearance of the images varies significantly across centers due to differences in staining protocols, imaging equipment, and tissue preparation procedures. Naively training a classifier on the source centers and evaluating on the target centers reveals a substantial performance gap, motivating the need for domain generalization strategies. To address this, we explore a variety of approaches, including data augmentation techniques, stain normalization methods, fine-tuning strategies and test-time adaptation methods. Together, these components form a pipeline designed to maximize generalization to unseen acquisition centers.

2. Architecture and Methodological Components

The Data. The dataset consists of histopathology image patches extracted from whole-slide images (WSIs) of either tumor (class 1) or non-tumor (class 0) tissue. The training set includes patches from three distinct medical centers, while the validation and test sets contain patches from two additional centers not seen during training. Figure 1 illustrates the distribution of samples across centers and classes, highlighting that the class distribution is balanced within each center but not across centers. We also provided in Figure 2 a

© 2026 CC-BY 4.0, J.-V. Martini & A.E. Kbadi.

UMAP visualization of the patch embeddings, which reveals which centers are more similar in terms of visual appearance and which ones are more distinct, further motivating the need for domain generalization techniques.

The Baseline. The baseline model is composed of a feature extractor and a linear probe. Frozen embeddings are extracted once from all images using the small version of the DINOv2, a Vision Transformer (VT) pretrained via self-supervised self-distillation on a large image corpus (Oquab et al., 2023). This design is computationally efficient and avoids overfitting, as gradient updates are confined to the lightweight linear classification head trained on top of the fixed representations. The preprocessing is very minimal, consisting of resizing the images using ImageNet statistics (Deng et al., 2009), consistent with DINOv2's pretraining regime. While this baseline achieves reasonable in-distribution performance, it provides no mechanism to handle the covariate shift introduced by unseen acquisition centers, motivating the components described in the following sections.

The Models. We retained DINOv2 (Oquab et al., 2023) as the main feature extractor, evaluating its small (`dinov2_vits14`), base (`dinov2_vitb14`), and large (`dinov2_vitl14`) variants to study the trade-off between capacity and computational cost. Beyond the frozen linear probe baseline, we applied a partial fine-tuning strategy in which the backbone is initially frozen, then the last N transformer blocks and final layer normalization are unfrozen and trained jointly with the classification head. We varied $N \in 2, 3, 5, 7, 10$, progressively allowing more of the backbone to adapt while preserving pretrained weights. The classification head is a single linear layer with a sigmoid output for binary prediction.

Convolutional architectures such as EfficientNet (Tan and Le, 2019) and ConvNeXt (Liu et al., 2022) were also considered as alternative backbones. EfficientNet provides a favorable accuracy-to-efficiency trade-off through compound scaling, while ConvNeXt modernizes ResNet-style architectures and achieves performance comparable to Vision Transformers on standard benchmarks. Both are suitable for patch-level classification and fine-tuning, but their supervised pretraining on ImageNet can reduce robustness under domain shift, as they are sensitive to low-level features like color and texture. In contrast, DINOv2 is trained using self-supervised learning on diverse data, encouraging the emergence of more invariant and transferable features (Oquab et al., 2023). This is particularly beneficial in the presence of strong covariate shift, such as staining and acquisition variability.

Data Augmentation for OOD Robustness. A key component of our approach to mitigate covariate shift is a stain-space augmentation strategy applied during training. Histopathology images from different acquisition centers differ substantially in their color appearance due to variations in staining protocols, reagent concentrations, and slide preparation procedures. To simulate this inter-center variability without requiring access to target-domain images, we implemented HED jittering, following the framework proposed by Tellez et al. (Tellez et al., 2019).

The augmentation operates by decomposing each RGB image into its Haematoxylin, Eosin, and DAB (HED) stain channels via color deconvolution (Ruifrok, 2001). Concretely, the RGB image is first converted to optical density (OD) space via $\mathrm{OD} = -\log(\mathbf{I})$, and then projected into HED space using the standard color deconvolution matrix. Random affine perturbations are then applied independently to each stain channel: $\tilde{h}_c = \alpha_c\, h_c + \beta_c, \quad c \in$

$\{\mathrm{H}, \mathrm{E}, \mathrm{D}\}$, where $\alpha_c \sim \mathcal{U}(1-\theta,\, 1+\theta)$ is a multiplicative scaling factor and $\beta_c \sim \mathcal{U}(-\theta,\, \theta)$ is an additive shift, both sampled independently per channel and per image. The perturbed HED representation is then projected back to OD space and converted to RGB via the inverse deconvolution matrix. We used $\theta = 0.05$, which corresponds to a subtle yet effective perturbation level.

In addition to stain jittering, the training pipeline includes standard spatial augmentations: random horizontal and vertical flips, and random rotations up to $\pm 90°$. At validation time, only resizing is applied.

Test-Time Augmentation. As a final component, we incorporate test-time augmentation (TTA) (Shanmugam et al., 2021) to improve robustness at evaluation time. TTA is not used during training or intermediate validation, but only applied post hoc once training is complete. For each input image, we generate geometric variants (horizontal and vertical flips, and 90° rotations), and average the predicted probabilities across these transformations to reduce variance and improve stability (Lyzhov et al., 2020). This approach mitigates sensitivity to orientation-specific features, which is particularly relevant in histopathology where tissue structures are largely orientation-invariant. We also experimented with adding stain perturbations (HED jittering) to TTA, but observed inconsistent gains relative to the increased computational cost.

For model selection, we report both standard validation accuracy and TTA-enhanced accuracy, and base the final choice on both metrics. This ensures that improvements are not solely driven by inference-time augmentation while still leveraging TTA as a practical performance booster.

3. Model Tuning and Comparison

3.1. Experimental Setup

3.2. Experiments and Results

4. Conclusion

Table 1: An Example Table

Dataset	**Result**
Data1	0.12345
Data2	0.67890
Data3	0.54321
Data4	0.09876

Algorithm 1: Random example

Input: $x_1, \ldots, x_n, w_1, \ldots, w_n$
Output: y, the net activation
$y \leftarrow 0$;
for $i \leftarrow 1$ **to** n **do**
  $y \leftarrow y + w_i * x_i$;
end

Acknowledgments

We thank a bunch of people.

References

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009.

Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11976–11986, 2022.

Alexander Lyzhov, Yuliya Molchanova, Arsenii Ashukha, Dmitry Molchanov, and Dmitry Vetrov. Greedy policy search: A simple baseline for learnable test-time augmentation. In *Conference on uncertainty in artificial intelligence*, pages 1308–1317. PMLR, 2020.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

Arnout Ruifrok. Quantification of histochemical staining by color deconvolution. *Analytical and quantitative cytology and histology/the International Academy of Cytology [and] American Society of Cytology*, 2001.

Divya Shanmugam, Davis Blalock, Guha Balakrishnan, and John Guttag. Better aggregation in test-time augmentation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 1214–1223, 2021.

Mingxing Tan and Quoc Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International conference on machine learning*, pages 6105–6114. PMLR, 2019.

David Tellez, Geert Litjens, Péter Bándi, Wouter Bulten, John-Melle Bokhorst, Francesco Ciompi, and Jeroen Van Der Laak. Quantifying the effects of data augmentation and stain color normalization in convolutional neural networks for computational pathology. *Medical image analysis*, 58:101544, 2019.

Appendix A. Details on the Data

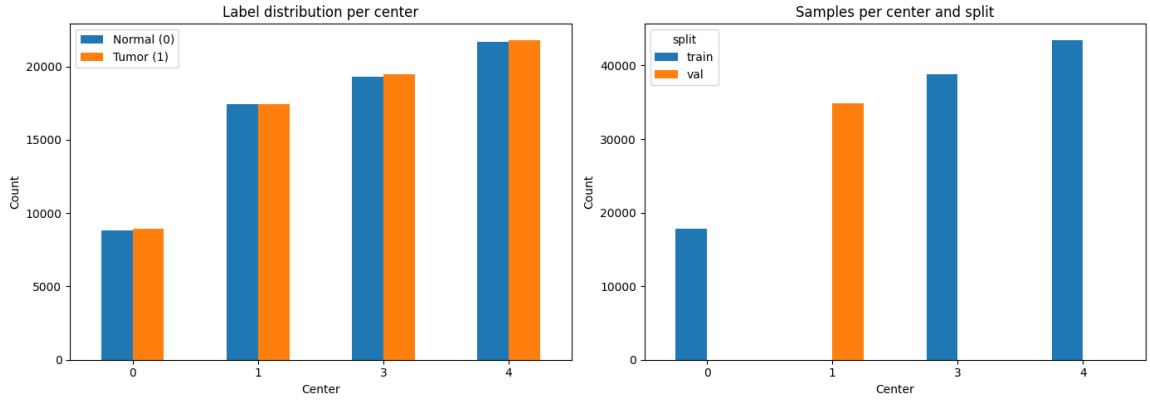

Figure 1: Distribution of samples across centers and classes.

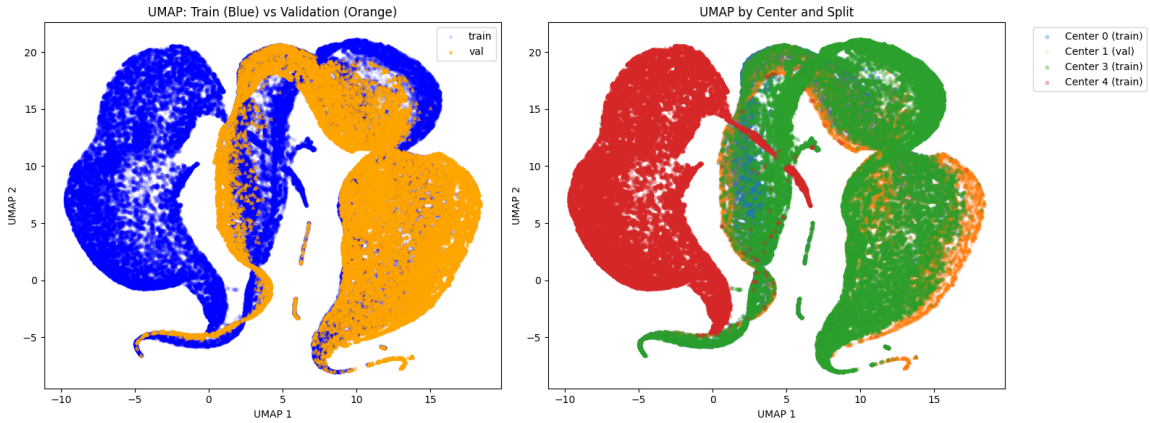

Figure 2: UMAP visualization of patch embeddings colored by **left:** training/validation split, and **right:** acquisition center.